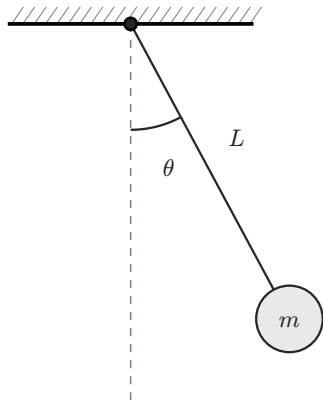
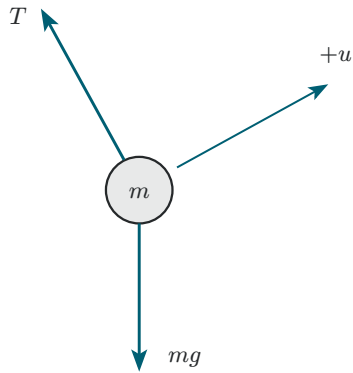


(a) Displaced pendulum



$u = L\theta$, measured along the arc

(b) Isolated bob



$$\sum F_u = -mg \sin(\theta)$$